

Different components of IoT



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***"Strength is Life, Weakness is Death."* – Swami Vivekananda**

IoT is not just a technology; it is an enormous ecosystem!

- **Community components:**

- ✓ **IoT Frameworks**

- Tools needed to design and implement IoT-based solutions and products

- ✓ **IoT Architectures**

- Graphical structure of the designed IoT-based solutions and products

- ✓ **IoT Core**

- Sensors & Actuators, microcontrollers, internet connectivity, service platform including security

- ✓ **IoT Gateway**

- It carries the responsibility to ensure bidirectional communication between IoT protocols and other networks

- ✓ **Cloud**

- Accepts, accumulates, maintains, stores, and process data in real time

- ✓ **Analytics**

- It indulges in conversion of data which results in meaningful suggestions and recommendations

- ✓ **User Interface / Visualization**

- Design sleek, visually appealing, interactive, and ease-of-use graphical user interface

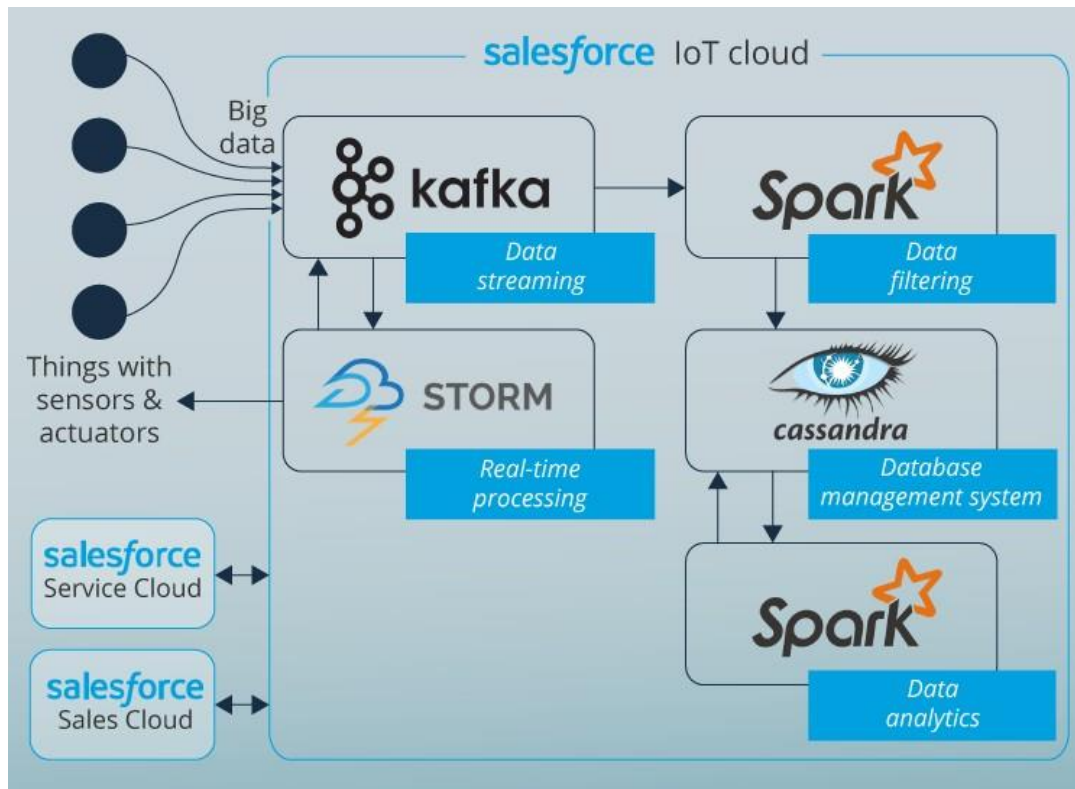
IoT Framework



- Framework provides a **development environment**.
 - It provides appropriate infrastructure **to design and implement the architecture**
- IoT framework comprises of **large number of components**
 - sensors, sensor systems, gateways, mobile app, embedded controller, data management platform, analytical platform, and so on.
 - support **interoperability** among all devices, provides **secure connectivity**, **reliability** in data transfer, **interface** to 3rd party application to built on it, and so on.

Few IoT Framework	Few IoT Framework
RTI (Real-Time Innovations) Connex DDS	Cisco Ultra IoT
Salesforce IoT cloud	Azure IoT
Eclipse IoT	ThingWorx
GE (General Electronic) Predix	AWS IoT
Watson IoT	Kaa

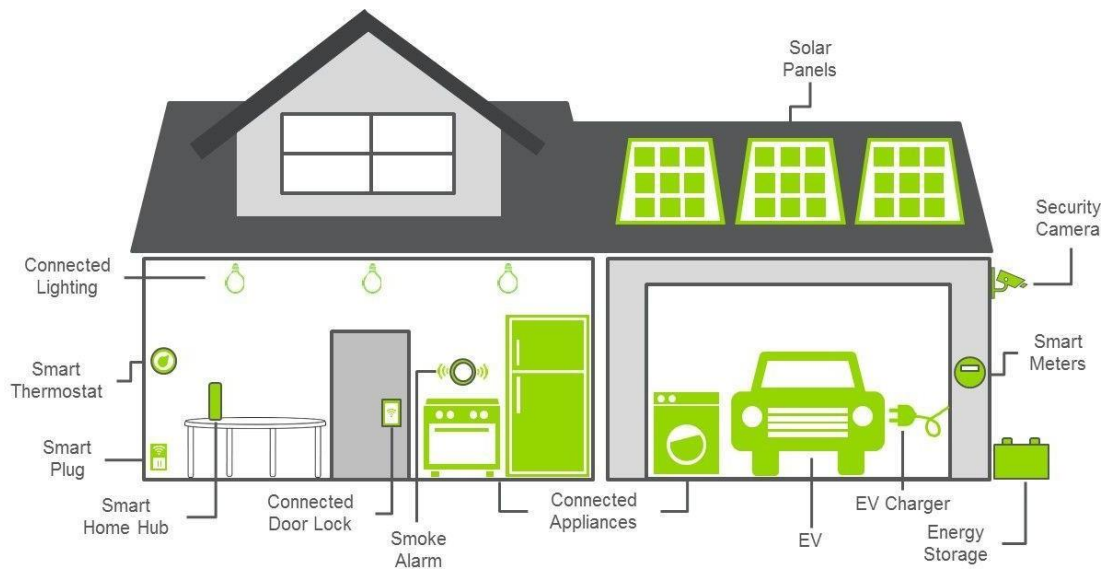
Salesforce IoT Cloud



- Salesforce IoT Framework
 - Combines **IoT** and **big data**
- A new data is sent to **Kafka**, where **Spark** is responsible to transmit it to **Cassandra** for further processing.
- Meanwhile, **Storm** analyses the dataset to formulate responses for current events
- The **backbone** of these technologies:
 - Salesforce's Heroku PaaS
 - AWS Cloud
 - Apache Technologies and Thunder event-processing engine

IoT Network Architecture

- At present, Network plays important role in business
- It should never be built without careful planning

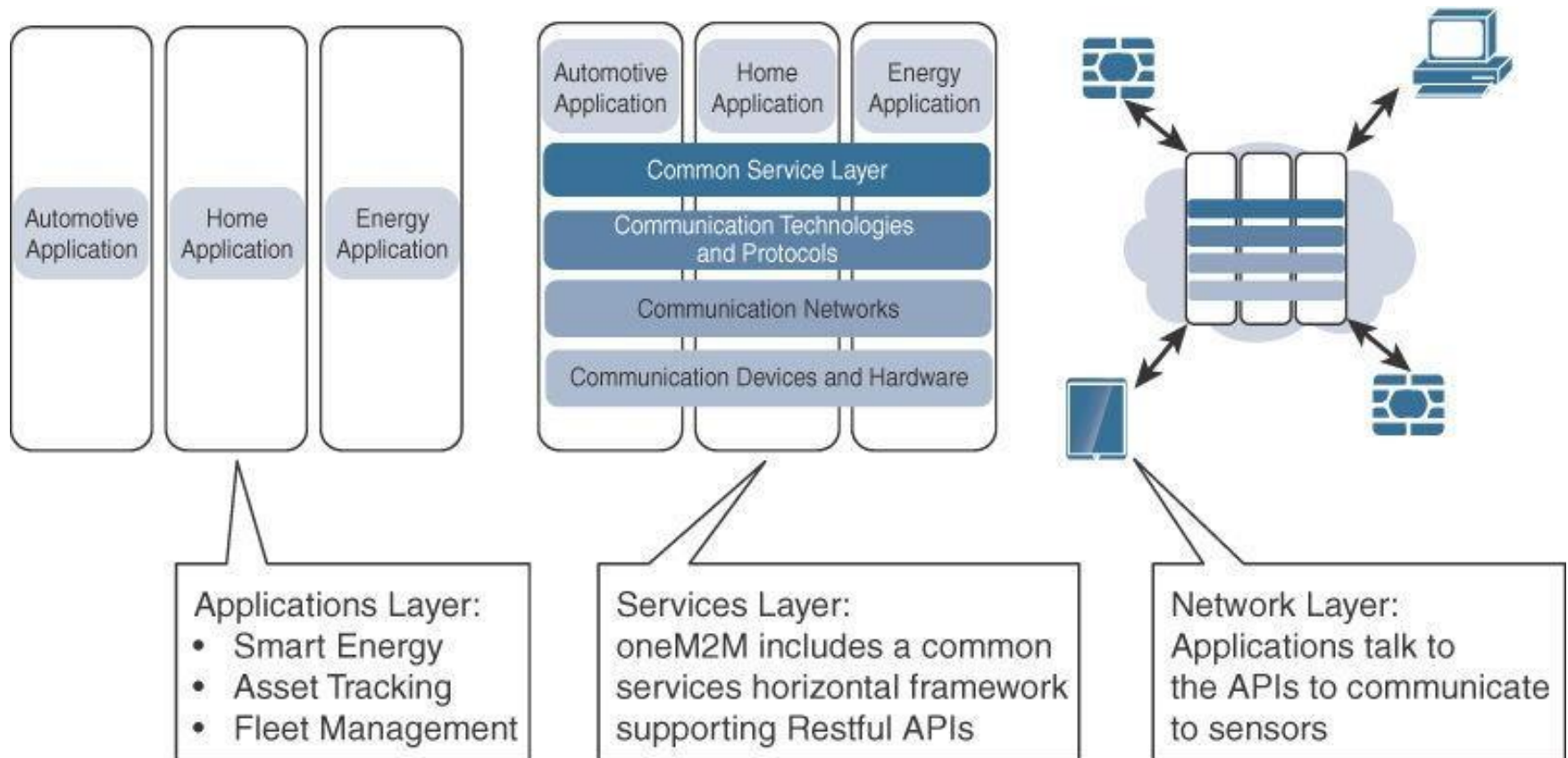


Driving forces:

- Scale
 - Security
 - Constrained devices
 - Massive data
 - Data analysis
 - Support to legacy devices
- Architecture is how you design (i.e. graphical structure) your application or solution.

oneM2M IoT Architecture

- Proposed by European Telecommunications Standards Institute (ETSI)
- Goal** : to create a **common services layer**, which can be readily **embedded in field devices** to allow communication with application servers.



IoTWF Architecture

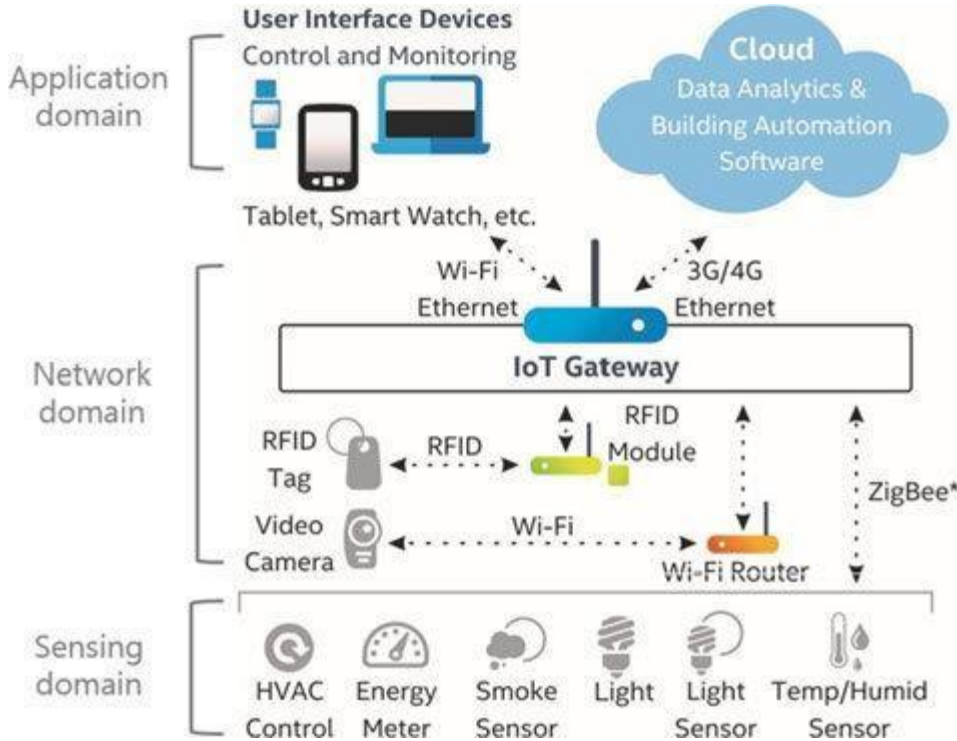
- IoTWF architectural committee (led by Cisco, IBM, Rockwell Automation, and others)
- Way of visualizing IoT from a technical / functional perspective

Levels

- 7 Collaboration & Processes**
(Involving People & Business Processes)
- 6 Application**
(Reporting, Analytics, Control)
- 5 Data Abstraction**
(Aggregation & Access)
- 4 Data Accumulation**
(Storage)
- 3 Edge Computing**
(Data Element Analysis & Transformation)
- 2 Connectivity**
(Communication & Processing Units)
- 1 Physical Devices & Controllers**
(The "Things" in IoT)

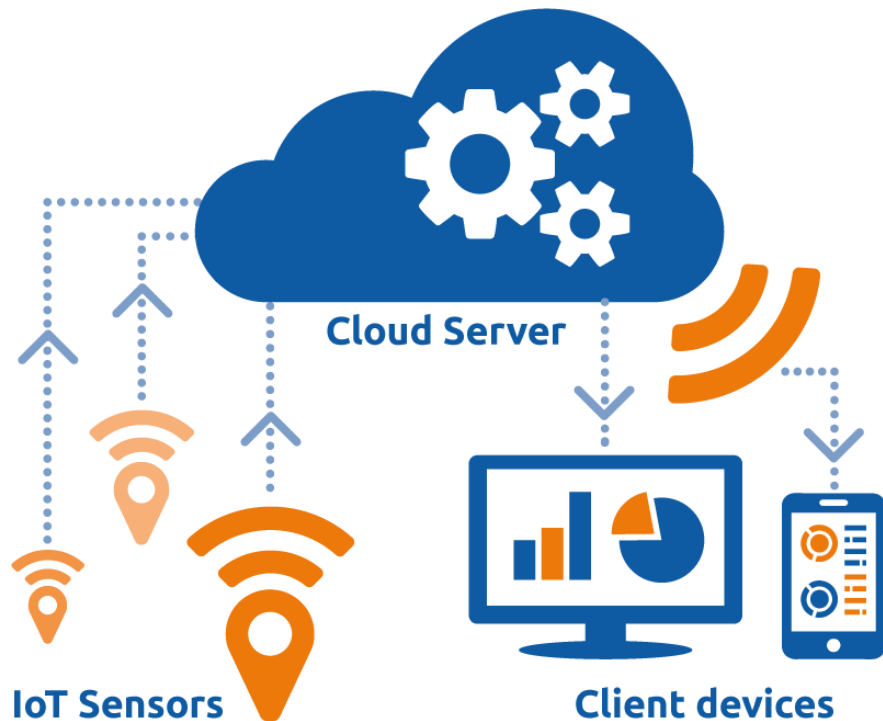


IoT Gateway



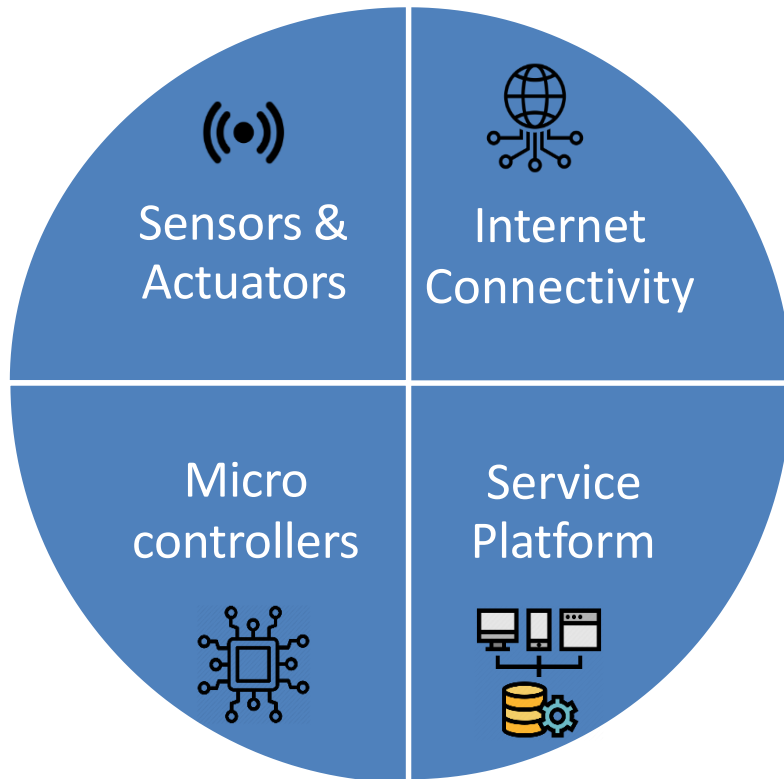
- It is a **physical device or software program** that serves as the connection point **between the two different types of networks**
- Provide bidirectional **communication**
 - Between IoT protocols and other networks
 - e.g. Zigbee <--> Ethernet
- Sometimes programmed to execute some **processing** operations
 - Edge computing
- It is necessary to maintain **security** to a certain extent
 - Can shield the entire IoT systems from any cyberattack

Source: B. Kang, D. Kim, H. Choo, "Internet of Everything: A Large-Scale Autonomic IoT Gateway", IEEE Transactions on Multi-scale Computing Systems, vol. 3, no. 3, 2017, pp. 206-214.



- IoT generates vast amount of Big Data;
- this in turn puts a huge strain on Internet Infrastructure.
- Cloud can facilitate to
 - Provide different services
 - Store huge amount of data
 - Process the data efficiently
- **Benefits of Cloud Platform in IoT**
 - Network Scalability
 - Data Mobility
 - Time to market
 - Security
 - Cost-effectiveness

Core Components of IoT



- **Sensors** - to gather data and events
- **Actuators** – responsible for moving and controlling a mechanism or system
- **Microcontrollers** - automatically controls sensors and actuators; makes them smart
- **Internet connectivity** – responsible for **sharing gathered information** and control command
- **Service Platform** – ability to deploy and manage the IoT devices and applications including **data management**, **data analytics** and all aspects of **security**

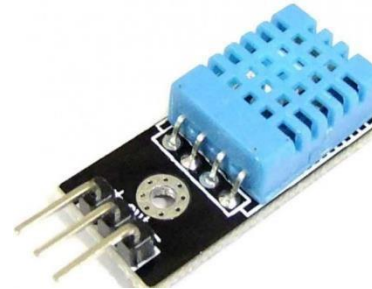
“Things” in IoT – Sensors



MQ135 - Air Quality Gas Sensor



Sound Detection Sensor



DHT11 - Temperature and Humidity Sensor



PIR Motion Detector Sensor



Pulse Sensor



LDR Light Sensor



Ultrasonic Distance Sensor



IR Sensor

“Things” in IoT – Actuators



4 Channel 5V Relay



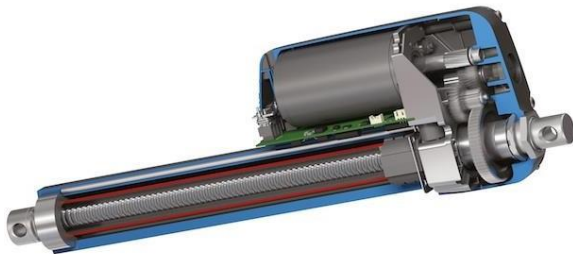
Servo Motor



DC Motor



Solenoid valve



Linear Actuators



LED



LCD

Connecting Smart Objects

Communication Criteria

- Range
- Frequency Bands
- Power Consumption
- Topology
- Constrained Devices
- Constrained-Node Networks

IoT Access Technologies



Comparison of Key Attributes



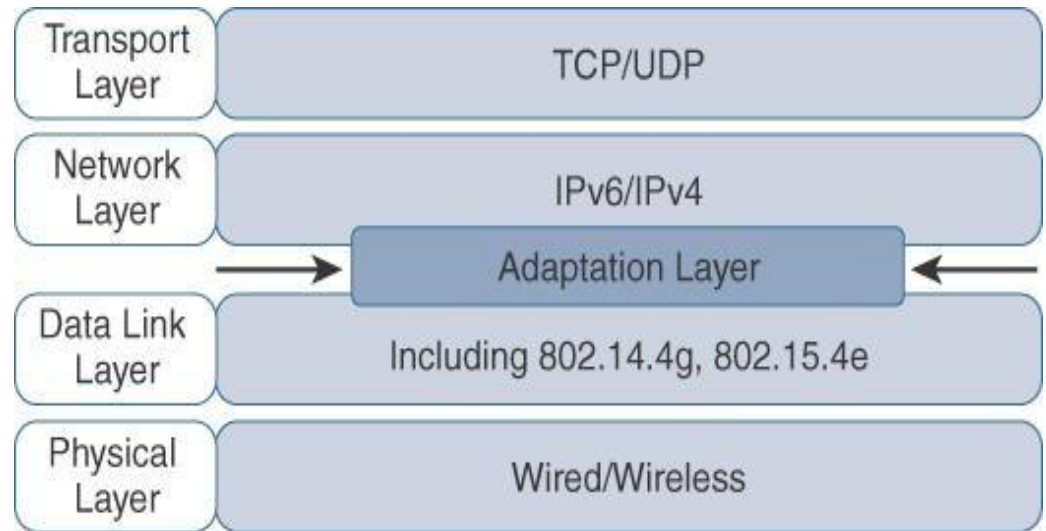
	Wi-Fi	BLE	Thread	Sub-GHz: TI	Sigfox	Zigbee	LoRa
Max. Data throughput	72 Mbps	2 Mbps	250 Kbps	200 Kbps	100 bps	250 Kbps	50 Kbps
Range	100 m	750 m	100 m	4 km	25 km	130 m	10 km
Topology	Star	P2P/ Mesh	Mesh/ Star	Star	Star	Mesh/ Star	Star of Star
Frequency	2.4 GHz	2.4 GHz	2.4 GHz	Sub-GHz	Sub-GHz	2.4 GHz	Sub-1GHz
Power consumption	1 Year (AA battery)	Up to years on a coin-cell battery for limited range					Few Years (AA battery)
IP at the device node	Yes	No	Yes	No	No	No	No
Deployed Devices	AP	smart phones	No	No	No	No	No

Source: Nick Lethaby “Wireless Connectivity for the IoT: one size does not fit all”, Texas Instruments, 2017

Utilizing IP for IoT

Key Advantages of IP

- Open and standard-based
- Versatile
- Ubiquitous
- Scalable
- Manageable
- Highly secure
- Stable and resilient



- IPv6 packets require a minimum MTU/PDU size of **1280 bytes**.
- The maximum size of a MAC layer frame in IEEE 802.15.4 is **127 bytes**.
 - It gives just **102 bytes for an IPv6 packet !!**

Need of packet/frame size optimization due to

- Constrained Nodes
- Constrained Networks

Modification in TCP/IP Stack

IP Protocol Stack

HTTP		RTP	
TCP	UDP		ICMP
IP			
Ethernet MAC			
Ethernet PHY			

Application

Transport

Network

Data Link

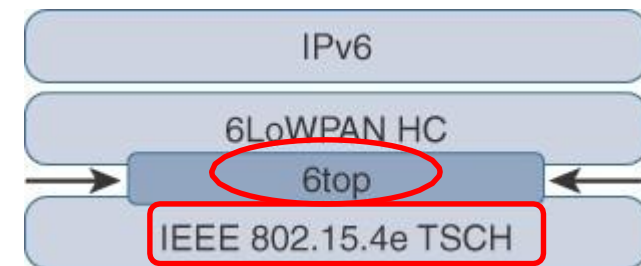
Physical

IoT Protocol Stack with 6LoWPAN Adaptation Layer

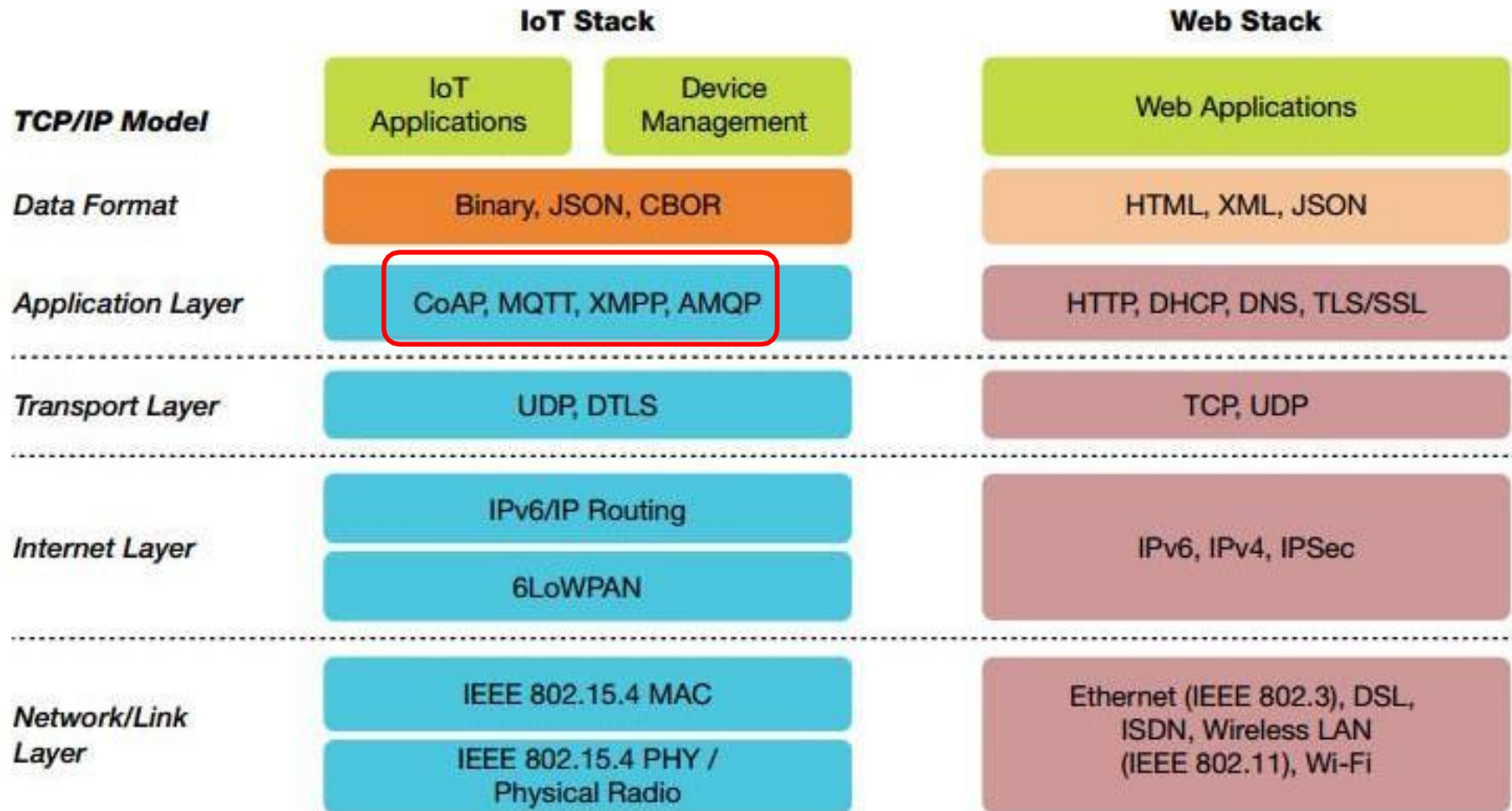
Application Protocols	
UDP	ICMP
IPv6	
LoWPAN	
IEEE 802.15.4 MAC	
IEEE 802.15.4 PHY	

- **WPAN**: Wireless Personal Area Networks
- **IEEE 802.15.4**: Low-rate WPAN
- **LoWPAN**: Low-Power WPAN
- **6LoWPAN**: IPv6 over LoWPAN
- **TSCH**: Time Synchronized Channel Hopping
- **6TiSCH**: IPv6 over the TSCH mode of IEEE 802.15.4e
- **6top**: 6TiSCH Operation Sublayer

In 6TiSCH IoT Network



Application Layer

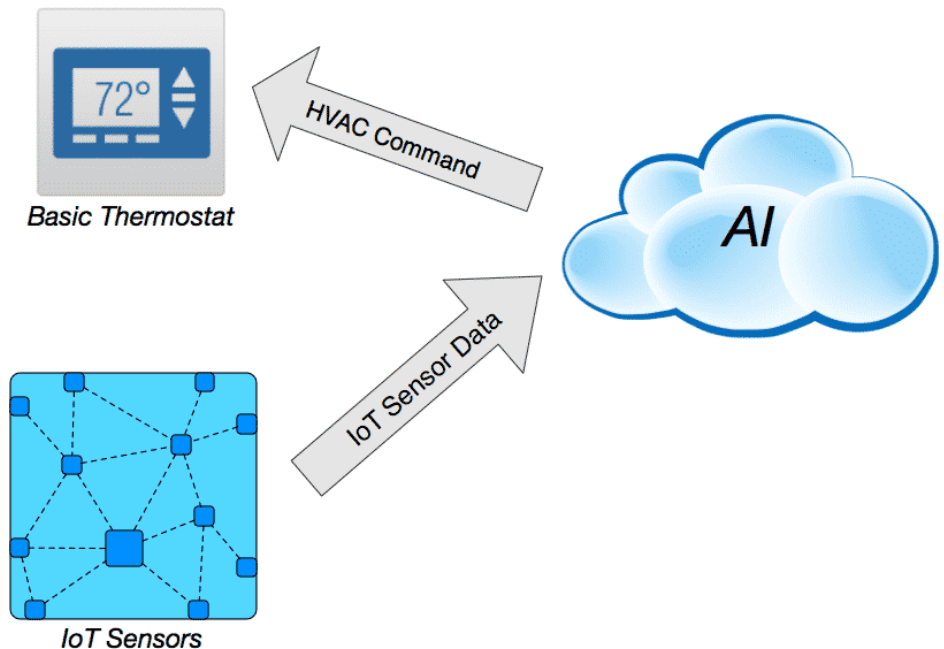


AI for IoT

- AI focuses on **putting human intelligence in machine**
- It gives the ability to a machine/program to **think and learn by itself**

Use of AI in IoT:

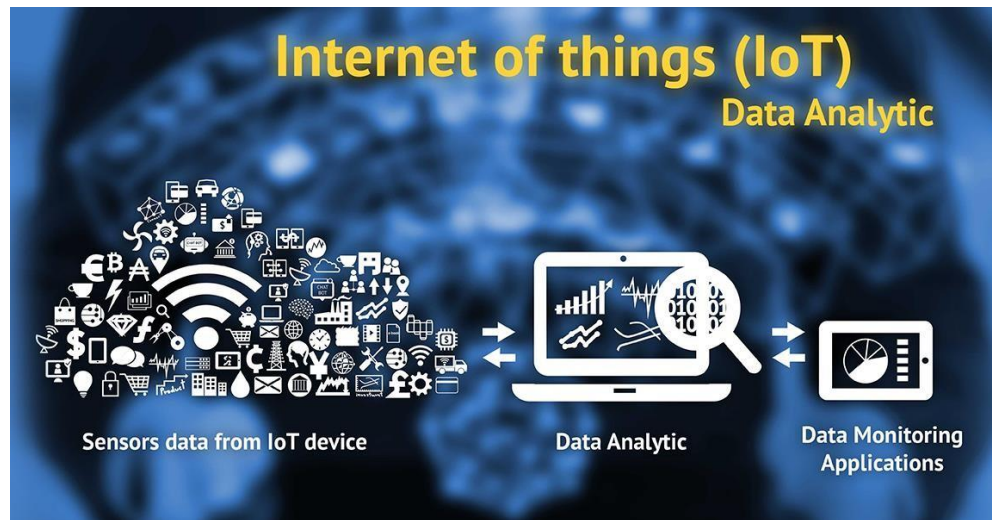
- **Smart Home**
 - Automated HVAC control
- **Industrial IoT**
 - Predictive maintenance
 - Optimized supply chain
- **Farming**
 - Smart farming
 - Interruption warning
- **Self-driving Car**
 - Mimic human driving on road
- **Health**
 - Auto-diagnosing any disease
 - Assistive healthcare



Data Analytics in IoT

“Data Analytics + IoT => Smart Business Solutions”

- ❖ The **business value of IoT** is not just in the ability to **connect devices**, but it comes from **understanding the data** these devices create.



Challenges:

- Huge **Volume**
 - Real-time data **flow**
 - **Variety** of data types
 - e.g. XML, video, SMS
 - Unstructured data
 - **Variable** data model and meaning / **value**
-
- **IoT analytics** is the **application of data analysis tools and procedures** to realize value from the huge volumes of data generated by connected IoT devices

Securing IoT

- Both the IoT **manufacturers** and their **customers** didn't care about the security !

Unauthorized access to IoT devices



Source: <https://www.theguardian.com/technology/2016/oct/26/ddos-attack-dyn-mirai-botnet>

Major cyber attack disrupts internet service across Europe and US;
October 26, 2016

Unauthorized access to IoT network



Source: <http://metropolitan.fi/entry/ddos-attack-halts-heating-in-finland-amidst-winter>

DDoS attack halts heating in Finland amidst winter;
November 7, 2016



Source: <https://www.youtube.com/watch?v=4oONdV5RYp8>

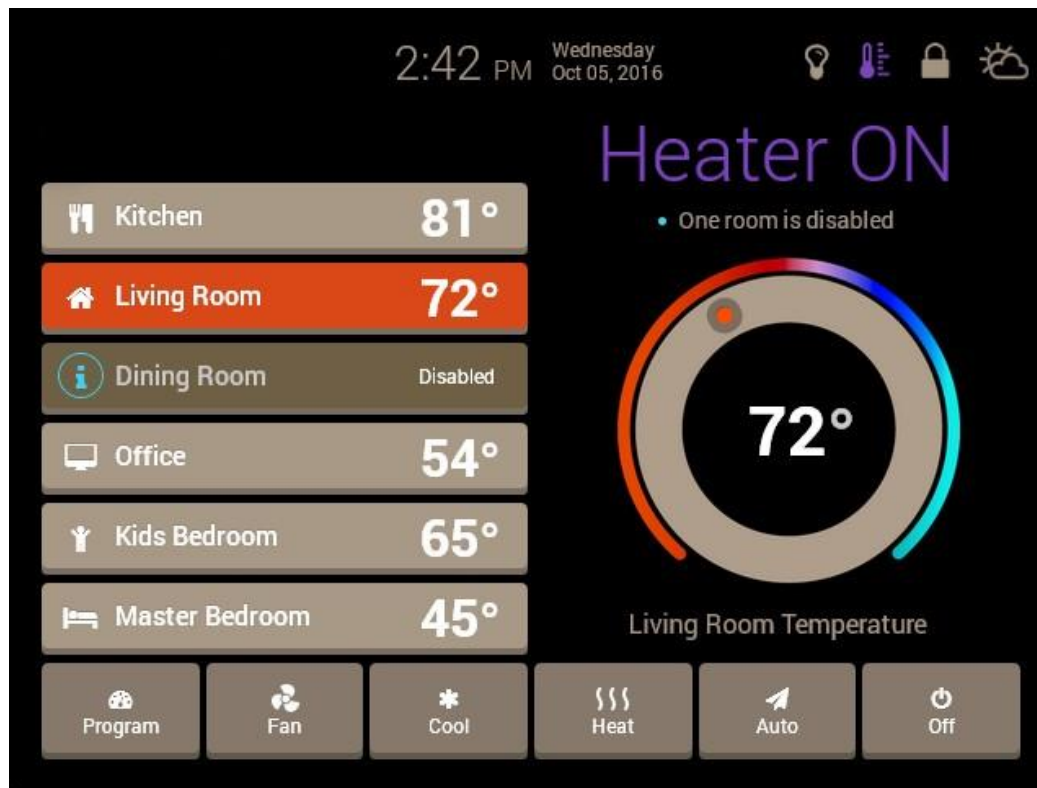
- US Military's Defense Advanced Research Projects Agency (DARPA) demonstrates hacking smart "Things"

Source: <https://www.youtube.com/watch?v=7E1WsdODxu0>



User Interface

- Information made available to the end-users
- Users can actively **check and act in** for their IoT system



Source: <https://data-flair.training/blogs/how-iot-works/>

Important Characteristics:

- ✓ Sleek design
- ✓ Visually appealing
- ✓ Interactive UI
- ✓ Ease-of-use
- ✓ Handy

Lessons Learned



- ✓ What is IoT Ecosystem
- ✓ What are different components
- ✓ What is IoT Framework
- ✓ Sensors & Actuators
- ✓ IoT Architecture
- ✓ IoT Access Technologies
- ✓ IoT Gateway
- ✓ Use of Cloud in IoT
- ✓ Data Analytics in IoT
- ✓ AI for IoT
- ✓ Security in IoT
- ✓ User Interface for IoT

Thanks!



Figures and slide materials are taken from the following Books:

1. David Hanes *et al.*, “**IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things**”, 1st Edition, 2018, Pearson India.
2. Mayur Ramgir, “**Internet of Things: Architecture, Implementation and Security**”, 1st Edition, 2020, Pearson India.